

Linux Fundamentals

BY

Prasad Nikam

Introduction

Linux is one of the most widely used operating systems in cybersecurity, particularly for servers, security tools, and penetration testing environments. Understanding how to interact with Linux through the command-line interface (CLI) is therefore an essential skill for anyone entering the cybersecurity field. This write-up covers the fundamentals of using a Linux terminal, including executing commands, navigating the file system, searching for files and text, and using shell operators. The objective is to build a strong foundation for working efficiently with Linux systems and understanding how commands can be combined to perform security-related tasks.

Talking to Linux

Linux is a popular choice for servers and the machines that you'll interact within cyber security. This interaction starts with the terminal, not a mouse. In this room, you will interact with your first ever Linux terminal - a place where we can run commands on the system.

Being familiar with the terminal (command line interface) on Linux is a critical skill as you will most likely spend most of your time here in Cyber security. From running hacking tools to hunting down attackers.

A command is an instruction that we can give the computer to perform a given task. One of the first commands we can do is `whoami`. This is important in Cyber security because you will often be changing users on the machine, which determines what you can and cannot do.

Whoami - tells you who you are on the system

When you run your first command, you will see some text. We call this output. Interacting with Linux feels like a conversation. You give it an instruction, and it gives you output. We can ask Linux to output specific text for us.

echo - output some specific text that is provided

To echo the text "TryHackMe", we can use `echo TryHackMe`. For multiple words, we will need to wrap these in quotes. For example, `echo "hello world"`. Using `echo` is just the start of a very important skill in Cyber security: being able create output and sending it somewhere. You will get to see how this skill is further used in the upcoming tasks.

Answer the questions below

What command would we use to find out who we are on the system?

✓ Correct Answer

What command would we use to output some text?

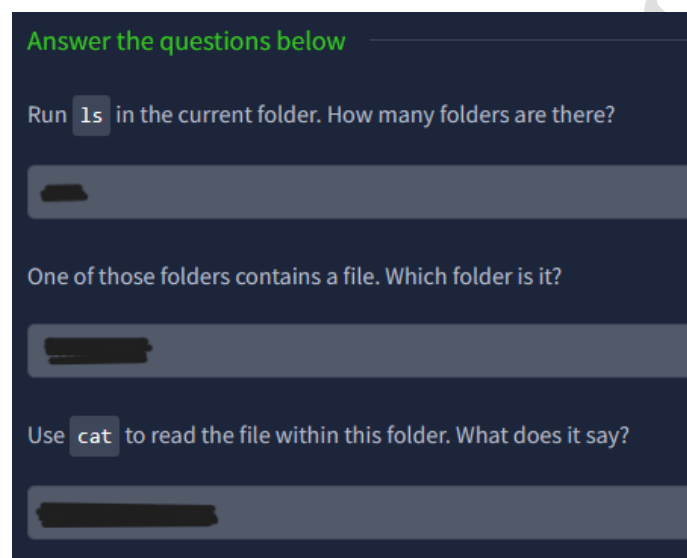
✓ Correct Answer

Finding your way around

Let's learn how to navigate around the system's files via the terminal. Four commands do almost all navigation

ls	list what's in the current folder
cd	change directory — move into a folder
cat	show the contents of a file
pwd	Print working directory — "where am I?"

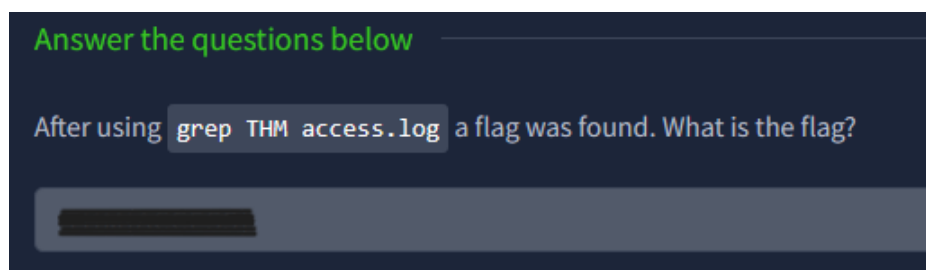
On Linux, you may notice that files and folders show as different colours. This makes it easy to identify what it is. On this Linux system, folders are blue.



Let the machine do the searching

We can use a set of Linux commands to help us efficiently search for and through files. Rather than scrolling through lots of text, we can have Linux do the hard work for us:

find	Search for files by their name. For example, find -name passwords.txt
grep	Searches <i>inside</i> for text. For example, grep "password123" passwords.txt

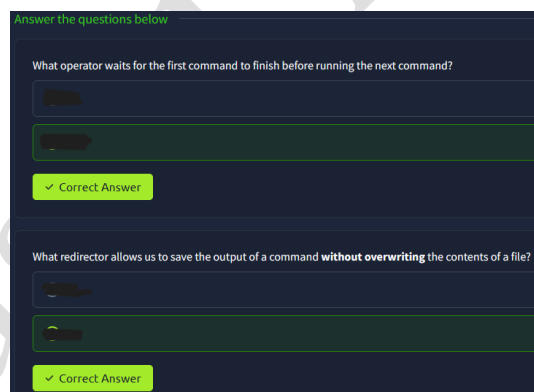


Shell operators

In Linux, there are a set of special characters that can combine commands together. These are called "Operators" which tell Linux how it should process both commands. From being able to combine commands to doing what's called a redirection - sending the output of commands elsewhere. Let's go over these now:

&	Runs the command, but does not wait for it to finish before you can do anything else. The command runs in the background, and is helpful for commands that might take a while to complete, or ones that you want to keep running.
&&	Runs both commands, but waits for the first command to finish first, before the next. Like a set of dominoes.
>	Used to redirect output. We can take the output of a command and send it to a file. This operator will overwrite anything that exists in the file.
>>	This redirector does the same thing, but instead of overwriting, it will just add the output to the bottom of the file.

For example, **sleep 10 &** runs the **sleep** command in the background while allowing you to continue using the terminal. **mkdir test && cd test** creates a **test** directory and moves into it only if the directory is created successfully. **echo hey > welcome** creates a file named **welcome** containing **hey**, overwriting any existing content. **echo hello >> welcome** adds **hello** to the end of the **welcome** file without overwriting its existing content.



Conclusion

This write-up introduced the fundamental concepts of interacting with Linux through the terminal. Commands such as **whoami**, **echo**, **ls**, **cd**, **cat**, and **pwd** provide the basic tools for identifying users, displaying information, and navigating the file system. Commands such as **find** and **grep** make it easier to locate files and search for specific information, while shell operators such as **&**, **&&**, **>**, and **>>** allow commands and their outputs to be managed efficiently. These fundamentals form an important foundation for further Linux and cybersecurity learning, where effective command-line usage is frequently required for system administration, investigation, and security testing.